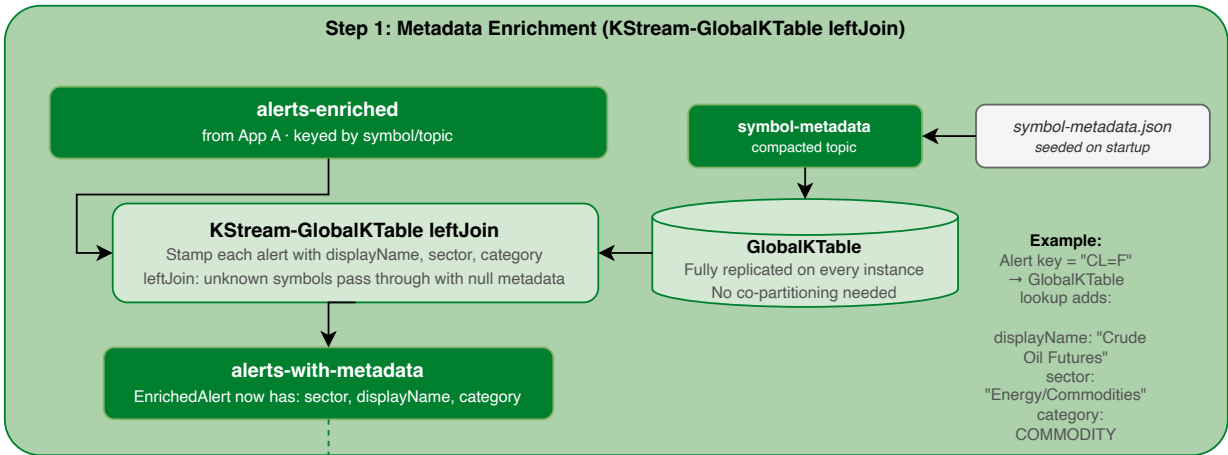


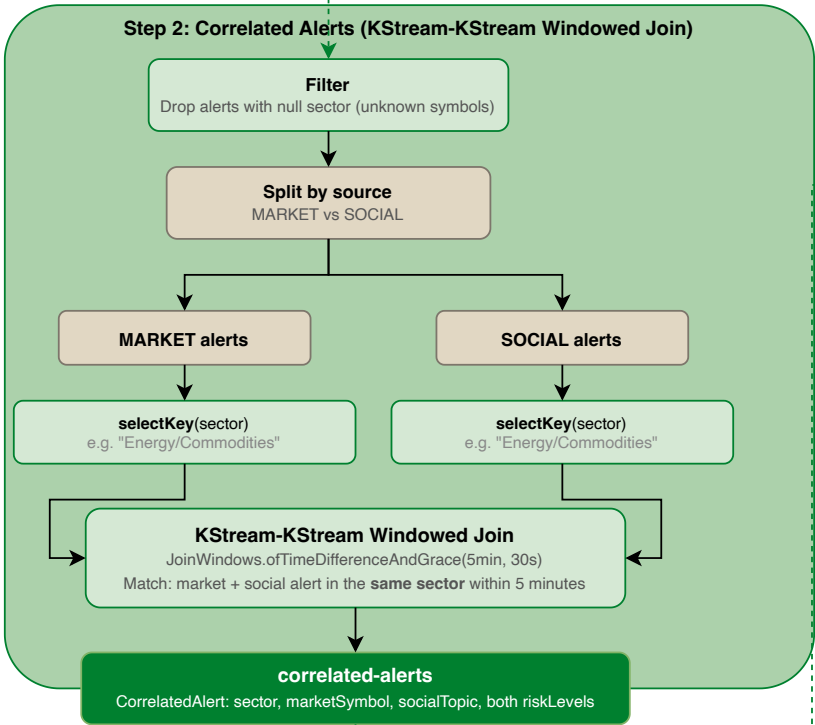
App B — NEWS Segment: Metadata Enrichment + Correlated Alerts + Summary

Step 1: Metadata Enrichment (KStream-GlobalKTable leftJoin)

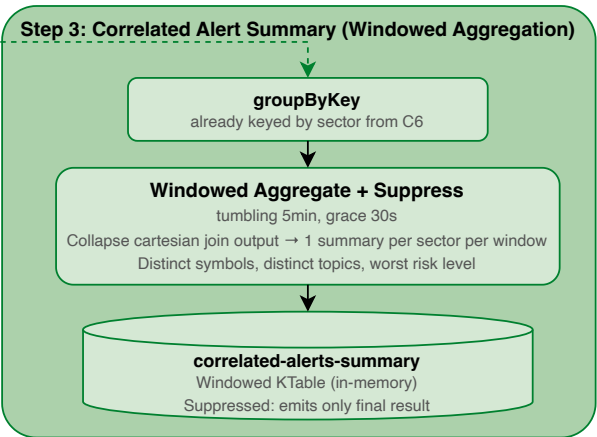


Step 1 is essential: The windowed join rekeys both streams by sector, but sector only exists after the GlobalKTable enrichment. Raw alerts have no sector field.

Step 2: Correlated Alerts (KStream-KStream Windowed Join)



Step 3: Correlated Alert Summary (Windowed Aggregation)



Suppressed Event Aggregator

Same pattern as signup-counts (C4b).
Emits one summary per sector when the 5-min window closes — no intermediate updates.